

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Medium

Question

Triangle FGH is similar to triangle JKL , where angle F corresponds to angle J and angles G and K are right angles. If $\sin(F) = \frac{308}{317}$, what is the value of $\sin(J)$?

Answer

- A. $\frac{75}{317}$
- B. $\frac{308}{317}$
- C. $\frac{317}{308}$
- D. $\frac{317}{75}$

Correct Answer: B

Rationale

Choice B is correct. If two triangles are similar, then their corresponding angles are congruent. It's given that right triangle FGH is similar to right triangle JKL and angle F corresponds to angle J . It follows that angle F is congruent to angle J and, therefore, the measure of angle F is equal to the measure of angle J . The sine ratios of angles of equal measure are equal. Since the measure of angle F is equal to the measure of angle J , $\sin(F) = \sin(J)$. It's given that $\sin(F) = \frac{308}{317}$. Therefore, $\sin(J)$ is $\frac{308}{317}$.

Choice A is incorrect. This is the value of $\cos(J)$, not the value of $\sin(J)$.

Choice C is incorrect. This is the reciprocal of the value of $\sin(J)$, not the value of $\sin(J)$.

Choice D is incorrect. This is the reciprocal of the value of $\cos(J)$, not the value of $\sin(J)$.